Problem 1

a)

M/M/1 queue
We first establish that the UCC can be modelled as an M/M/1 queue. This requires iid exponentially distributed interarrival and service times, one server, and independence between the arrival and service processes.
Since patients arrive according to a Poisson process with rate λ, the interarrival times are iid Exp(λ). The treatment times are independent and exponentially distributed with mean $1/\mu$, so they are iid Exp(μ). Furthermore, the UCC has one server, and treatment times are independent of arrivals. Thus, all conditions for an M/M/1 queue are satisfied.

Birth-and-death process.
Having established the M/M/1 model, we can describe the number of patients in the UCC as a birth-and-death process. Let $X(t)$ denote the number of patients in the system at time t. The independent Poisson arrivals and memoryless exponential service times imply the Markov property.
An arrival increases $X(t)$ by one, while a completed treatment decreases it by one. Using the small-time properties of the Poisson and exponential distributions, the transition probabilities are

$$P(X(t+h) = i+1 \mid X(t) = i) = \lambda h + o(h), \quad i \geq 0,$$
$$P(X(t+h) = i-1 \mid X(t) = i) = \mu h + o(h), \quad i \geq 1.$$

Dividing by h and letting $h \to 0^+$ gives the birth and death rates:

$$\boxed{\lambda_i = \lambda \ (i \geq 0), \qquad \mu_0 = 0, \qquad \mu_i = \mu \ (i \geq 1).}$$

Expected time in the UCC.
We now use these birth-and-death rates to determine the expected time a patient spends in the UCC. This is the expected sojourn time $E[S]$, which includes both waiting time and treatment time.
To apply Little's law, we first need the long-run average number of patients in the system. We therefore begin by finding the limiting distribution.
For the birth-and-death process, we have

$$\theta_0 = 1, \qquad \theta_k = \frac{\lambda_0 \cdots \lambda_{k-1}}{\mu_1 \cdots \mu_k} = \left(\frac{\lambda}{\mu}\right)^k.$$

Since $\lambda < \mu$ is given, the corresponding geometric series converges:

$$\sum_{k=0}^{\infty} \theta_k = \frac{1}{1 - \lambda/\mu}.$$

Normalizing the weights gives the limiting distribution

$$\pi_k = \frac{\theta_k}{\sum_{j=0}^{\infty} \theta_j} = \left(1 - \frac{\lambda}{\mu}\right)\left(\frac{\lambda}{\mu}\right)^k, \quad k \geq 0.$$

We can now use this distribution to calculate the expected number of patients. Using that a geometric random variable Y taking values $k = 0, 1, 2, \ldots$ has probability mass function

$$P(Y = k) = (1-p)^k p.$$

Comparing this with π_k, we recognize the limiting distribution as geometric with $p = 1 - \lambda/\mu$.
The long-run average number of patients in the UCC is therefore

$$L = E[Y] = \frac{1-p}{p} = \frac{\lambda}{\mu - \lambda}.$$

Finally, having found the average number of patients, we can apply Little's law, which relates the average number in the system to the arrival rate and the expected sojourn time:

$$L = \lambda E[S].$$

Solving for $E[S]$ and substituting our expression for L gives

$$\boxed{E[S] = \frac{L}{\lambda} = \frac{1}{\lambda}\left(\frac{\lambda}{\mu - \lambda}\right) = \frac{1}{\mu - \lambda}.}$$

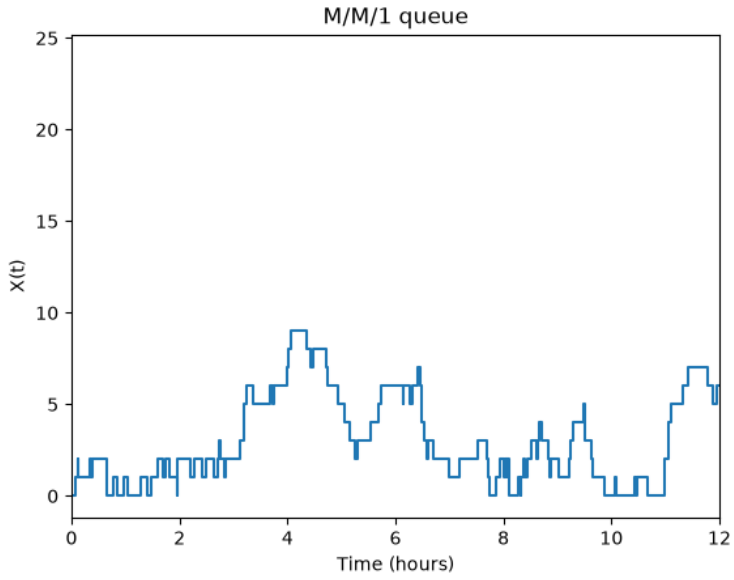
Thus, the expected time a patient spends in the UCC, including both waiting and treatment, is $1/(\mu - \lambda)$.

b) Simulation of the M/M/1 queue

We simulated the birth-and-death process $X(t)$ over 50 days, using $\lambda = 5$ and $\mu = 6$ patients per hour. Since arrivals follow a Poisson process, the interarrival times were simulated independently from $\text{Exp}(\lambda)$. Treatment times were generated independently from $\text{Exp}(\mu)$, also independent of arrivals.
For each patient i, treatment begins at the later of their arrival time A_i and the previous patient's departure time D_{i-1}. Thus, with treatment duration S_i, departures were calculated recursively as

$$D_i = \max(A_i, D_{i-1}) + S_i, \qquad D_0 = 0.$$

Arrivals and departures were then combined chronologically to construct $X(t)$, increasing by one at arrivals and decreasing by one at departures.
Figure 1 shows one realization during the first 12 hours. The process remains constant between events.



Figur 1: One realization of $X(t)$ during the first 12 hours.

Estimation of the expected time in the UCC. The long-run mean number of patients was estimated using the time-weighted average

$$\hat{L} = \frac{1}{T}\sum_{i=0}^{n-1} X(t_i)(t_{i+1} - t_i).$$

By Little's law, the expected time in the UCC was estimated as $\widehat{E[S]} = \hat{L}/\lambda$.
The simulation was repeated independently 30 times, giving a mean estimate of

$$\widehat{E[S]} = 0.9860 \text{ hours.}$$

An approximate 95% confidence interval was calculated using Python's `stats.sem()` and `stats.t.interval()`, giving

$$CI_{95\%} = [0.9411,\ 1.0310] \text{ hours.}$$

From part a), the theoretical expected time is

$$E[S] = \frac{1}{\mu - \lambda} = \frac{1}{6-5} = 1 \text{ hour.}$$

The theoretical value lies within the confidence interval, and the simulated estimate is close to the exact value, indicating good agreement with the M/M/1 model.

c)

M/M/1 queue
The process $\{U(t) : t \geq 0\}$ can be modelled as an M/M/1 queue. Each arriving patient is independently classified as urgent with probability p. By Poisson thinning, urgent arrivals form a Poisson process with rate $p\lambda$, giving i.i.d. Exp($p\lambda$) interarrival times.
As in part a), treatment times are i.i.d. Exp(μ) and independent of arrivals. The UCC has one server, and urgent patients always have priority, immediately interrupting treatment of normal patients. Consequently, normal patients do not affect the service of urgent patients, so the urgent queue behaves as if it has its own server.
Thus, all conditions for an M/M/1 queue are satisfied.

Arrival rate
The arrival rate is the expected number of arrivals per unit time. Let $A_U(t)$ denote the number of urgent arrivals up to time t. Since urgent arrivals form a Poisson process with rate $p\lambda$, we have

$$A_U(t) \sim \text{Poisson}(p\lambda t).$$

Thus, the arrival rate is

$$\lambda_U = \frac{E[A_U(t)]}{t} = \frac{p\lambda t}{t} = \boxed{p\lambda}.$$

Long-run mean number of urgent patients.
As in part a), we use the birth-and-death process to determine the limiting distribution, now with arrival rate $p\lambda$. The weights are

$$\theta_0 = 1, \qquad \theta_k = \frac{\lambda_0 \cdots \lambda_{k-1}}{\mu_1 \cdots \mu_k} = \left(\frac{p\lambda}{\mu}\right)^k.$$

Since $0 < p < 1$ and $\lambda < \mu$, we have $p\lambda < \mu$, and therefore

$$\sum_{k=0}^{\infty} \theta_k = \frac{1}{1 - p\lambda/\mu}.$$

Normalizing the weights gives

$$\pi_k = \left(1 - \frac{p\lambda}{\mu}\right)\left(\frac{p\lambda}{\mu}\right)^k, \qquad k \geq 0.$$

This is a geometric distribution with parameter $q = 1 - p\lambda/\mu$. Thus, the long-run mean number of urgent patients is

$$E[U] = \frac{1-q}{q} = \frac{p\lambda/\mu}{1-p\lambda/\mu} = \boxed{\frac{p\lambda}{\mu - p\lambda}}.$$

d)

Not M/M/1 queue
The process $\{N(t) : t \geq 0\}$ does not behave as an M/M/1 queue because normal patients can only receive treatment when no urgent patients are present.
For example, if $N(t)$ transitions from 2 to 1, a normal patient has completed treatment, meaning the remaining patient can immediately begin treatment. However, if $N(t)$ transitions from 0 to 1, the arriving patient may have to wait if urgent patients are present.
Thus, even when $N(t) > 0$, the server may be occupied treating an urgent patient. The departure rate of normal patients is therefore μ when $U(t) = 0$, but 0 when $U(t) > 0$. Hence, the departure rate cannot be determined from $N(t)$ alone, unlike an M/M/1 queue, where the departure rate is always μ whenever the system is nonempty.

Long-run mean number of normal patients
We have that the total amount of patients is normal + urgent, meaning

$$X = U + N,$$

if we take the expected value of this, and using linearity and moving to the other side, we obtain

$$E[X] = E[U + N] \implies E[N] = E[X] - E[U].$$

Using the results from 1a and 1c we get

$$\begin{aligned} E[N] &= E[X] - E[U] \\ &= \frac{\lambda}{\mu - \lambda} - \frac{p\lambda}{\mu - p\lambda} \\ &= \frac{\lambda(\mu - p\lambda) - p\lambda(\mu - \lambda)}{(\mu - \lambda)(\mu - p\lambda)} \\ &= \boxed{\frac{\lambda\mu(1-p)}{(\mu - \lambda)(\mu - p\lambda)}}. \end{aligned}$$

e)

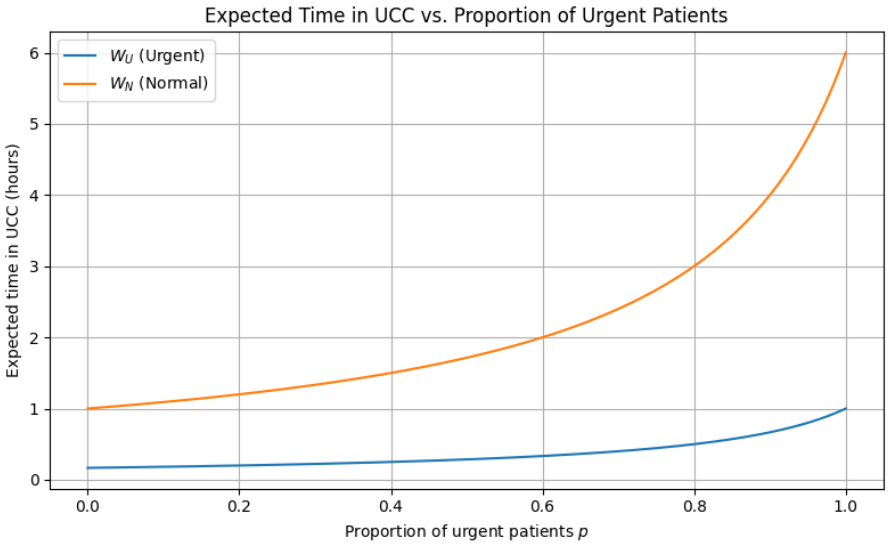
Expected time urgent patient
We can use Little's law to find

$$W_U = \frac{L_U}{\lambda_U} = \frac{E[U]}{p\lambda} = \frac{p\lambda}{\mu - p\lambda}\frac{1}{p\lambda} = \frac{1}{\mu - p\lambda}.$$

Expected time normal patient
We can use Little's law to find

$$W_N = \frac{L_N}{\lambda_N} = \frac{E[N]}{(1-p)\lambda} = \frac{\lambda\mu(1-p)}{(\mu - \lambda)(\mu - p\lambda)}\frac{1}{(1-p)\lambda} = \frac{\mu}{(\mu - \lambda)(\mu - p\lambda)}.$$

f)



Figur 2: Expected time in the UCC for urgent and normal patients as a function of p, with $\lambda = 5$ and $\mu = 6$.

Interpretation of the results.
As p increases, the proportion of urgent patients increases. Consequently, urgent patients experience longer waiting times due to more urgent arrivals, while normal patients experience even longer waiting times because urgent patients always have priority.
When $p \approx 0$, almost all patients are normal, meaning that normal patients rarely get interrupted. The expected time for normal patients approaches that of a standard M/M/1 queue.
When $p \approx 1$, almost all patients are urgent. Normal patients are frequently interrupted and must wait until all urgent patients have been treated, resulting in significantly longer times in the UCC.

Expected time for normal patients.
Using $\lambda = 5$ and $\mu = 6$, we have

$$W_N(p) = \frac{\mu}{(\mu - \lambda)(\mu - p\lambda)} = \frac{6}{6 - 5p}.$$

For $p \to 0$, we obtain

$$\lim_{p \to 0} W_N(p) = \frac{6}{6 - 0} = 1 \text{ hour}.$$

For $p \to 1$, we obtain

$$\lim_{p \to 1} W_N(p) = \frac{6}{6 - 5} = 6 \text{ hours}.$$

Thus, the expected time for normal patients increases from 1 hour to 6 hours as the proportion of urgent patients approaches 1.

Finding p when $W_N = 2$ hours.
We solve

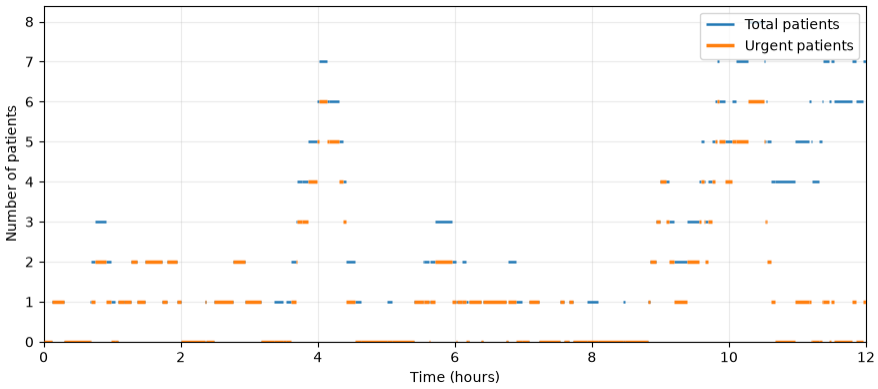
$$W_N(p) = 2 \implies \frac{6}{6 - 5p} = 2 \implies 6 = 12 - 10p \implies 10p = 6 \implies p = 0.6$$

Therefore, when 60% of arriving patients are urgent, the expected time a normal patient spends in the UCC is 2 hours.

g)

Expected time urgent patient

We plot the expected waiting times for urgent and normal patients as functions of the proportion of urgent patients p, using $\lambda = 5$ and $\mu = 6$.



Figur 3: Simulated total number of patients and number of urgent patients in the UCC during the first 12 hours, with $\lambda = 5$, $\mu = 6$, and $p = 0.80$.

As p increases, the expected time spent at the UCC increases for both patient groups, but much more for normal patients due to the priority system. As $p \to 1$, almost all patients are urgent, and the expected times approach 1 hour for urgent patients and 6 hours for the few remaining normal patients.

We also calculate 95% confidence intervals based on the 30 simulations to see how reliable our estimates are. The theoretical values lie within these intervals, which suggests that our simulations give reasonable results.

Problem 2: Calibrating Climate Models

We model the unknown deterministic function $y(\theta)$ using a Gaussian process

$$Y \sim \mathcal{GP}(\mu, C),$$

where $\mu(\theta)$ is the mean function and $C(\theta_1, \theta_2)$ is the covariance function. We say that $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$,

$$\mathbf{Y}(\boldsymbol{\theta}) \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma), \qquad \Sigma_{ij} = C(\theta_i, \theta_j).$$

We find $C(\theta_1, \theta_2)$ trough

$$\begin{aligned} C(\theta_1, \theta_2) &= \sigma_1 \sigma_2 \operatorname{Corr}(Y(\theta_1), Y(\theta_2)) \\ &= 0.5^2 (1 + 15|\theta_1 - \theta_2|) e^{-15|\theta_1 - \theta_2|}. \end{aligned}$$

a) Prediction

The expectation and variance are given by

$$E[Y(\theta)] = 0.5, \qquad \operatorname{Var}[Y(\theta)] = 0.5^2.$$

To calculate the conditional distribution we first construct the covariance matrix explained above. Since the grid values $\mathbf{Y}_g$ and observations $\mathbf{Y}_o$ are jointly Gaussian, the conditional distribution is

$$\mathbf{Y}_g \mid \mathbf{Y}_o = \mathbf{y}_o \sim \mathcal{N}(\boldsymbol{\mu}_c, \Sigma_c),$$

where

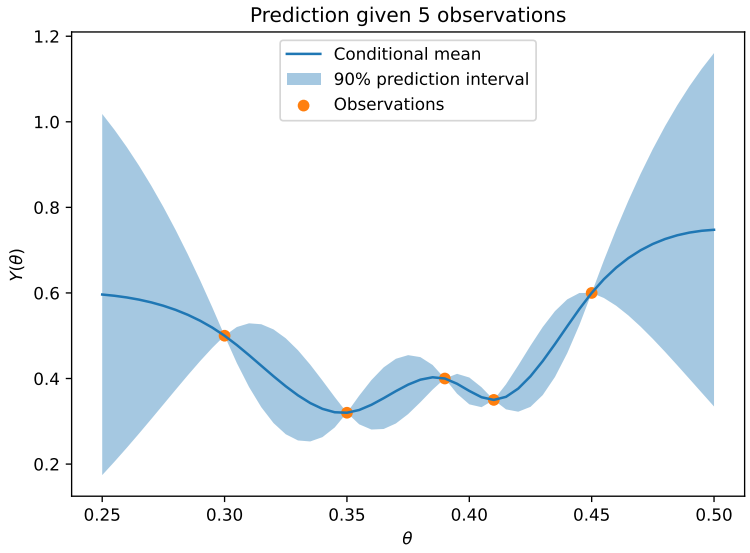
$$\boldsymbol{\mu}_c = \boldsymbol{\mu}_g + K_{go}K_{oo}^{-1}(\mathbf{y}_o - \boldsymbol{\mu}_o),$$
$$\Sigma_c = K_{gg} - K_{go}K_{oo}^{-1}K_{og}.$$

Here, K_{gg}, K_{oo} and K_{go} are the covariance matrices between grid points, observations, and grid-observation pairs, respectively. The dimentions are

$$K_{gg} \in \mathbb{R}^{51\times 51},$$
$$K_{oo} \in \mathbb{R}^{5\times 5},$$
$$K_{go} \in \mathbb{R}^{51\times 5},$$
$$K_{og} \in \mathbb{R}^{5\times 51}.$$

These expressions give the predicted mean and covariance based on the five observations.
The conditional mean is plotted against 51 equally spaced grid points on the interval $\theta \in [0.25, 0.5]$, with the 90% prediction interval given by

$$\mu_{c,i} \pm \Phi^{-1}(0.95)\sqrt{\Sigma_{c,ii}}$$



Figur 4: Conditional mean and 90% prediction interval of $Y(\theta)$ given the five observations. The uncertainty decreases near observed parameter values and increases further away.

b) Probability of Reaching the Target

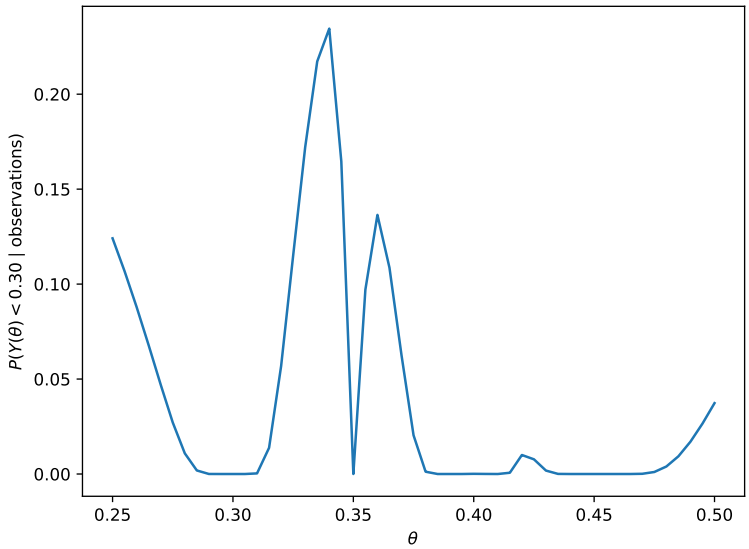
Using the conditional mean and variance from (a), we calculate the probability of achieving $Y(\theta) < 0.30$ using the Gaussian cumulative distribution function Φ:

$$P(Y(\theta) < 0.30 \mid \mathbf{y}_o) = \Phi\left(\frac{0.30 - \mu_c(\theta)}{\sqrt{\Sigma_c(\theta,\theta)}}\right).$$

We evaluate this probability at each of the 51 grid points and plot it as a function of θ. Higher probabilities indicate parameter values more likely to achieve the target score. At the observed points, the probability is zero since all observed scores exceed 0.30.

c) A New Observation

Here we repeat the procedure from (a) and (b), now including the additional observation. The covariance matrices and conditional distribution are recalculated using all six observations, giving updated predictions and probabilities of achieving $Y(\theta) < 0.30$.

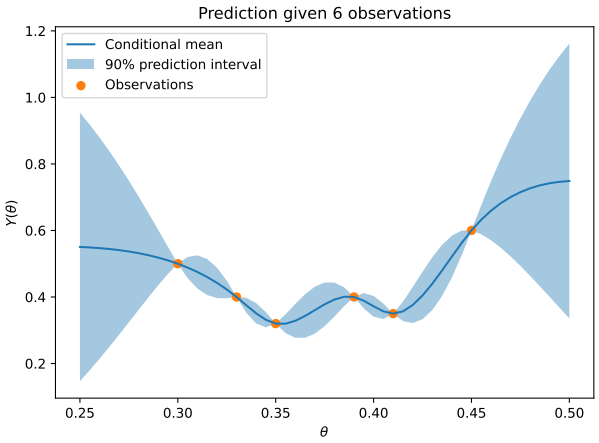


Figur 5: Conditional probability of achieving a score below 0.30 as a function of θ, given the five observations. Higher values indicate a greater probability of reaching the target.

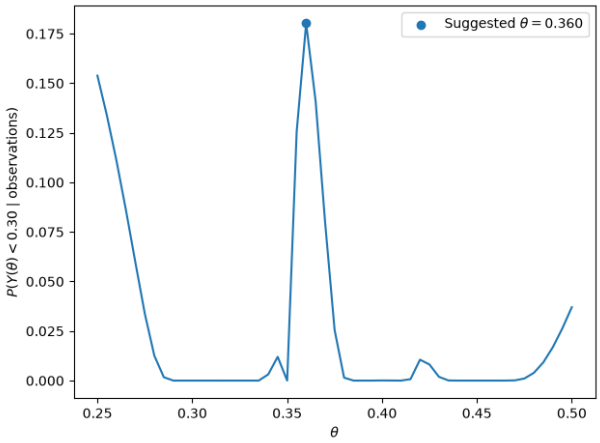
If we were to repeat the experiment, we should choose the parameter θ that maximizes the probability of achieving the target. Thus, the optimal parameter is

$$\boxed{\theta^* = \underset{\theta \in [0.25, 0.50]}{\arg\max} \; P(Y(\theta) < 0.30 \mid \mathbf{y}_o)}$$

We evaluate this probability at all the grid points and select the one with the highest probability.



Figur 6: Conditional mean and 90% prediction interval, given the 6 observations



Figur 7: Conditional probability of achieving a score below 0.30 as a function of θ, given all six observations.

From the updated probability distribution, the highest probability is obtained at $\theta^* = 0.360$, with a probability of 0.180. The additional observation changes the predicted probabilities by providing more information about the unknown function.

Division of work:

Elias Veines-Jensen wrote the solutions to Problem 1 a)–c), Michael Birkeland wrote the solutions to Problem 1 d)–f), Jakob Westgård wrote the solution to Problem 1 g), and Thomas Paulen wrote the solutions to Problem 2. All group members contributed to the review of the project.